

Forecasting Daily Henry Hub Natural Gas Spot Prices

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Introduction

The volatility of natural gas prices has profound implications for energy markets, economic planning, and environmental policies. As the benchmark for US natural gas prices, the Henry Hub spot price is pivotal for guiding decisions across stakeholders, including energy producers, industrial consumers, and policymakers. Accurate price forecasting is critical for mitigating risks, optimizing resource allocation, and implementing effective regulatory interventions. However, forecasting natural gas prices presents inherent challenges due to the complex interplay of factors such as market fundamentals, geopolitical tensions, and weather-related disruptions.

The urgency to develop reliable forecasting models has intensified amidst global energy transitions and economic uncertainties. Developments such as the shale gas revolution, the decoupling of oil and gas prices, and the growing integration of renewable energy sources have added layers of complexity to natural gas markets. Additionally, the COVID-19 pandemic underscored the susceptibility of energy markets to unexpected disruptions, resulting in unprecedented volatility and reinforcing the need for robust predictive tools.

This study tackles these challenges by creating a comprehensive forecasting framework for Henry Hub natural gas spot prices. Utilizing historical data on natural gas and crude oil prices, it applies advanced time series and forecasting techniques to enhance prediction accuracy. The findings aim to provide actionable insights for energy market participants, from optimizing production schedules to informing policy decisions. By addressing existing gaps in the literature, this research contributes to the evolving field of energy price forecasting and supports more resilient decision-making in an increasingly dynamic market landscape.

Literature Review

Accurate forecasting of natural gas prices has been a critical focus of research due to its significance in ensuring energy market stability and informing policymaking. Traditional econometric models, such as ARIMA and GARCH, have been widely used to capture historical patterns and price volatility. For instance, Salehnia et al. (2011) analyzed nonlinear models, including local linear regression and artificial neural networks (ANNs), finding that while dynamic local linear regression outperformed ANNs, its predictions were often noisy due to inherent market complexities. Similarly, Serletis and Shahmoradi (2012) employed GARCH models to study Henry Hub price volatility, emphasizing the influence of weather conditions and capacity constraints on price fluctuations.

The emergence of machine learning has added new dimensions to natural gas price forecasting. Recent studies have highlighted techniques like support vector regression (SVR) and ensemble learning for their improved predictive accuracy. Ceperic et al. (2017) compared classical time series methods with machine learning models, demonstrating that feature selection significantly enhances forecasting by reducing subjective inputs. Jana and Ghosh (2021) further advanced this area with residual-driven ensemble models and wavelet transformations, which accounted for data granularity and improved accuracy during volatile periods, such as the COVID-19 pandemic.

Despite these advancements, forecasting challenges persist, particularly in addressing structural shifts in the natural gas market, such as the decoupling of oil and gas prices and the effects of shale gas production. Recent advancements in forecasting methodologies highlight the importance of incorporating external market dynamics and unconventional techniques to better capture the complexities of natural gas price behavior. For example, Wong-Parodi, Lekov, and Dale (2005) compared the predictive accuracy of the Energy Information Administration's (EIA) Annual Energy Outlook with Henry Hub futures prices, finding that the latter outperformed due to its responsiveness to short-term market fluctuations. Similarly, Anderson et al. (2011) stressed the importance of integrating consumer sentiment and market psychology into forecasting models, which enhances the ability to predict price trends, especially during periods of uncertainty. Gao et al. (2020) underscored the importance of flexible time-varying parameter models for capturing these dynamics in rapidly evolving markets. Additionally, Farhangdoost et al. (2020) advocated for composite forecasting approaches that combine individual model outputs, improving accuracy during periods of heightened volatility. These studies collectively emphasize the potential of hybrid approaches that combine econometric methods, machine learning, and behavioral insights to account for the diverse factors influencing natural gas prices.

While progress has been made, gaps remain in integrating external factors like geopolitical events, regulatory changes, and technological innovations into predictive models. This study seeks to bridge these gaps by incorporating crude oil price dynamics with Henry Hub natural gas prices, utilizing advanced time series and forecasting techniques to enhance prediction accuracy.

Data

The dataset consists of monthly Henry Hub Natural Gas Spot Prices and Cushing West Texas Intermediate (WTI) Crude Oil prices, sourced from the U.S. Energy Information Administration (EIA). Spanning January 1997 to October 17, 2024, the dataset includes 334 months of data, offering a robust foundation for developing reliable time series models and accurate forecasts for natural gas prices.

Table 1: Variable Names, Definitions, and Units

Variables	Definition	Unit
Month	Month and Corresponding Years	Index
NG	Henry Hub Natural Gas Spot Prices	Current USD per Barrel
CO	WTI Crude Oil Spot Prices	Current USD per Million Btu

The data processing and time series forecasting are conducted using R. Descriptive statistics reveal that natural gas prices have a mean of \$4.132 and a median of \$3.440 per million Btu, while crude oil prices exhibit a mean of \$58.71 and a median of \$58.23 per barrel (Table 2).

This research is highly relevant to various stakeholders. Energy companies can utilize price forecasts to optimize production schedules, aligning output with anticipated price increases to maximize profitability. Commodity traders can develop informed strategies for buying or selling futures contracts, effectively hedging risks and capitalizing on price fluctuations. Policymakers may rely on these forecasts to guide decisions on strategic reserves or implement market-stabilizing measures during periods of volatility. Similarly, industrial consumers, including manufacturers and large-scale gas users, can use forecasts to secure favorable supply contracts or adjust consumption patterns, potentially reducing long-term costs by stockpiling gas in anticipation of price hikes.

Results and Analysis

This research focuses on forecasting monthly Natural Gas (NG) prices by exploring their temporal dynamics. Using advanced time series analysis, the study identifies significant components, applies transformations to achieve stationarity, and employs robust forecasting models, including ETS and ARIMA, while testing the integration of Crude Oil (CO) prices through dynamic regression.

The time series plot (Figure 1) highlights the volatility of NG prices compared to CO prices, suggesting sensitivity to market shocks, demand fluctuations, and seasonality. Notable price spikes in NG warrant further investigation into the temporal structure. Descriptive statistics (Table 2) show NG prices have pronounced variability, ranging from 1.49 to 13.42, compared to CO's relatively stable range of 11.35 to 133.88, emphasizing the need for seasonality and trend modeling.

The relationship between NG and CO prices is further examined in Figure 2, which includes scatterplots, histograms, and correlation coefficients. The weak correlation coefficient, significant at the 1% level, suggests limited linear dependence but potential nonlinear interactions, justifying CO's inclusion in dynamic regression for further exploration.

Next, logarithmic transformation of NG prices (Figure 3) here fails to mitigate volatility due to the absence of exponential patterns. On the other hand, the subseries plot (Figure 4)

reveals recurring seasonal patterns, suggesting seasonality as a significant driver of NG price dynamics. The fluctuating bar heights across the months further confirm the presence of seasonality in the data.

Classical decomposition (Figure 5) isolates the underlying trend as a major contributor to fluctuations, though residuals show patterns during initial periods. ACF diagnostics (Figure 6) confirm residuals are not white noise, necessitating further transformation. Similar results emerge from X11 (Figures 7, 8) and STL decomposition (Figures 9, 10), where Ljung-Box (lb) test results (p-value = 0) indicate significant autocorrelation in residuals.

A 3-month moving average (Figure 11) smooths short-term fluctuations, revealing NG prices are primarily driven by medium- to long-term trends. The smoothed series (Figure 12) overlaps closely with the original, highlighting minimal short-term irregularities. Linear regression decomposition (Figure 13) isolates a statistically significant negative trend, while the first difference (Figure 14) tries to remove both trend and seasonality. However, the ACF diagnostics and Ljung-Box tests (p-values of 0 and 0.0197, respectively) show significant patterns in residuals, indicating these methods do not fully capture the series dynamics.

A Trend-Seasonal Linear Model (TSLM) is then applied to simultaneously estimate trend and seasonal components, finding a statistically significant negative trend (at 1% level) coefficient of -0.004152. Regression on seasonal dummies (Figures 15, 16) and seasonal differencing (Figures 17, 18) similarly fail to remove all patterns, confirmed by residual diagnostics and lb p-values (0). Subsequently, a first difference was applied to the NG price series to remove the overall trend, followed by seasonal differencing to eliminate recurring seasonal patterns by subtracting values from the same period in the previous year in figure 19 and 20. This also led to the same conclusions as before with significant autocorrelation patterns and a lb p-value < 1%, rejecting the null hypothesis that the residuals are white noise and the data is stationary.

With the results discussed above, forecasting began with exponential smoothing first. R automatically selected the ETS (M, M, N) model based on the lowest AIC (1568.254), AICc (1568.327), and BIC (1579.687) compared to the other ETS models. The associated error terms are presented in Table 3. The multiplicative error structure effectively models NG price variability. However, the model generates a flat forecast (Figure 21) with significant autocorrelation (lb p-value = 0.006), indicating insufficient stationarity and unreliable forecasts.

Eventually, the ARIMA (5, 1, 0) (0, 0, 1) [12] model was manually chosen to address trend and seasonality components of the series aptly and produce a better forecast. This model outperformed ETS based on lower AIC (768.93 vs. 1568.25), AICc (769.28 vs. 1568.327), and BIC (795.59 vs. 1579.687) values (Table 4) and a higher lb p-value (0.254 vs. 0.006), ensuring stationarity. Table 5 outlines ARIMA coefficients, highlighting the significance of higher-

order AR terms and the seasonal moving average term (SMA1) in capturing short-term and seasonal dependencies. The residual diagnostics (Figure 22) also support the model's robustness, with no significant autocorrelation and residuals closely following the zero line. The histogram of residuals also shows symmetry, further validating the model's adequacy.

The ARIMA model's predictive capability is evaluated by forecasting NG prices for the test set (2023 onward) given the training data (before 2023) and comparing these predictions with observed values (Figure 23). The close alignment between the predicted and observed data confirms the model's effectiveness in capturing the series dynamics. Figure 24 extends this analysis with a 12-month forecast, showing logical continuation of historical trends and fluctuations, with Table 6 presenting the corresponding forecasted values. The forecasted line accurately mirrors the underlying pattern, demonstrating the model's reliability for future predictions. Interestingly, the forecasted price for November 2024, \$2.12 per million Btu, matches exactly with the actual price reported by the EIA, underscoring the accuracy of the forecasting model.

To explore additional explanatory power, a dynamic regression model integrating CO prices was tested. However, as shown in Table 7 and Figure 25, the results remain consistent with the basic ARIMA model, suggesting that CO prices contribute little to improving the model's performance. Ultimately, the ARIMA (5, 1, 0) (0, 0, 1) [12] model emerges as the most effective and, thereby, the final model for forecasting NG prices, leveraging its ability to address trend and seasonality comprehensively and produce reliable predictions for the next 12 months where it projects a slight increase in prices, followed by mostly a stable plateau. This model can also be mathematically represented as below using backshift operators.

Here, the ARIMA (5, 1, 0) (0, 0, 1) [12] model combines both trend and seasonal components where $(p, d, q) = (5, 1, 0)$; $(P, D, Q) = (0, 0, 1)$; and $m = 12$. The trend part includes five autoregressive (AR) terms, first-order differencing, and no moving average (MA) terms. The seasonal ARIMA (0, 0, 1) [12] component includes no seasonal AR terms, no seasonal differencing, and one seasonal MA term with a periodicity of 12 representing 12 months.

For the trend component ARIMA (5, 1, 0), the AR terms are $(1 - \phi_1 B - \phi_2 B^2 - \phi_3 B^3 - \phi_4 B^4 - \phi_5 B^5)$, where $\phi_1, \phi_2, \phi_3, \phi_4$, and ϕ_5 are the autoregressive coefficients, and B is the backshift operator. The first-order differencing is represented as $(1 - B)$. Therefore, the non-seasonal part of the model can be written as: $(1 - \phi_1 B - \phi_2 B^2 - \phi_3 B^3 - \phi_4 B^4 - \phi_5 B^5) (1 - B) y_t$.

For the seasonal component ARIMA (0, 0, 1) [12], the seasonal MA term is expressed as $(1 + \theta_1 B^{12})$, where θ_1 is the seasonal MA coefficient and B^{12} represents the backshift operator applied 12 steps back. Since there is no seasonal differencing or seasonal AR term, the seasonal component is solely represented as: $(1 + \theta_1 B^{12}) \varepsilon_t$.

Combining both components, the ARIMA (5, 1, 0) (0, 0, 1) [12] model is expressed as: $(1 - \phi_1 B - \phi_2 B^2 - \phi_3 B^3 - \phi_4 B^4 - \phi_5 B^5) (1 - B) y_t = (1 + \theta_1 B^{12}) \varepsilon_t$.

Conclusion

This study offers a robust framework for forecasting Henry Hub Natural Gas Spot Prices, emphasizing the effectiveness of traditional time series models in volatile markets. The ARIMA (5, 1, 0) (0, 0, 1) [12] model emerged as the most reliable, effectively capturing trends and seasonality while accounting for the inherent variability of natural gas prices. Incorporating crude oil prices through dynamic regression yielded limited additional insights, indicating that natural gas price dynamics are largely driven by strong seasonality and autocorrelation. These findings align with existing literature, reaffirming the utility of ARIMA models while acknowledging the challenges posed by non-linear relationships and external market influences.

The study's implications span diverse energy market stakeholders. Accurate price forecasting is crucial for mitigating risks, optimizing production schedules, and guiding strategic policies in volatile markets. Energy producers can leverage these insights to refine operational strategies, while policymakers and commodity traders gain tools to navigate supply chain uncertainties and price fluctuations. Future research could enhance this framework by incorporating machine learning models to better capture non-linear dependencies and structural market shifts. Expanding the analysis to global or high-frequency datasets could further deepen understanding of cross-market dynamics and broaden the applicability of predictive models in an increasingly interconnected energy landscape. In brief, this study provides valuable contributions to energy price forecasting and supports the pursuit of sustainable energy transitions in an era of heightened market complexity bridging theoretical modeling and practical applications.

Reference

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10. U.S. Energy Information Administration. (2024). Cushing, OK: WTI Crude Oil Spot price. <https://www.eia.gov/dnav/ng/hist/rngwhhdD.htm>

Appendix

Figure 1: Time Series of Natural Gas (NG) and Crude Oil prices (CO)

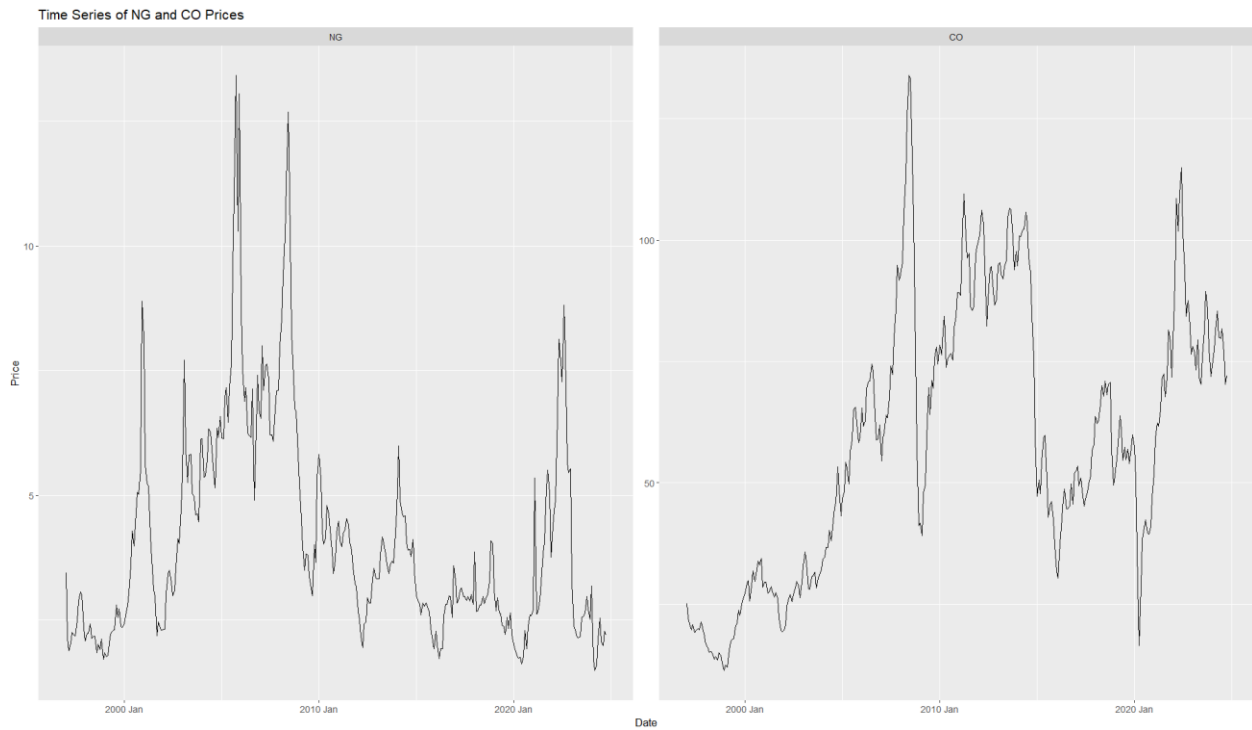


Figure 2: Scatter Plots, Histograms, and Correlation Coefficients of NG and CO Prices

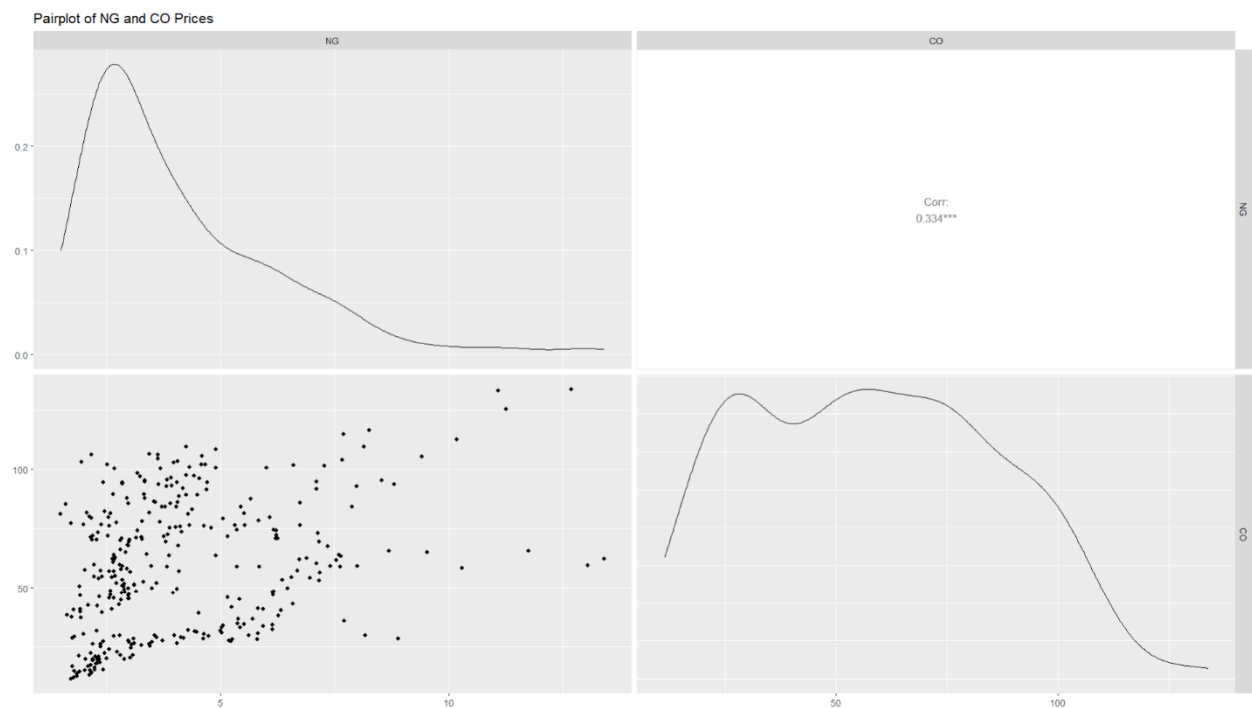


Table 2: Summary Statistics of NG and CO Monthly Prices

	Min	1 st Quartile	Median	Mean	1 st Quartile	Max
NG	1.490	2.583	3.440	4.132	5.255	13.420
CO	11.35	32.99	58.23	58.71	79.69	133.88

Figure 3: Time Series of Original and Log Transformed NG Prices

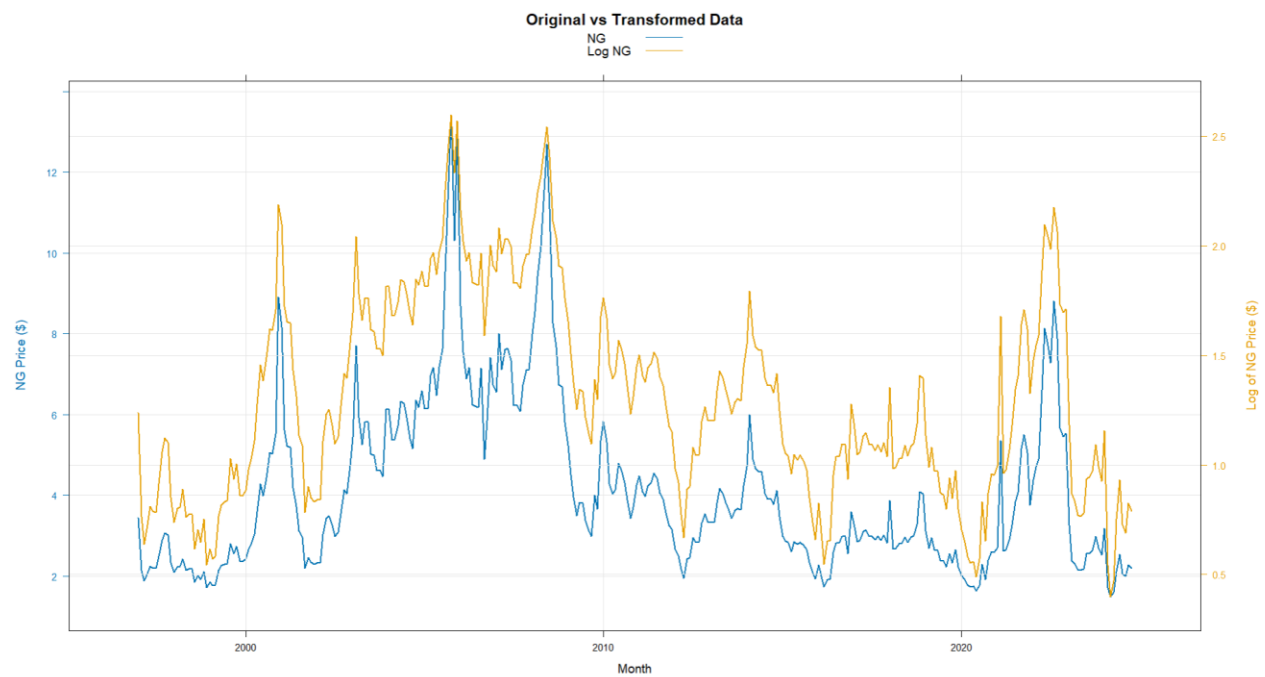


Figure 4: Monthly Natural Gas Prices Subseries Plot

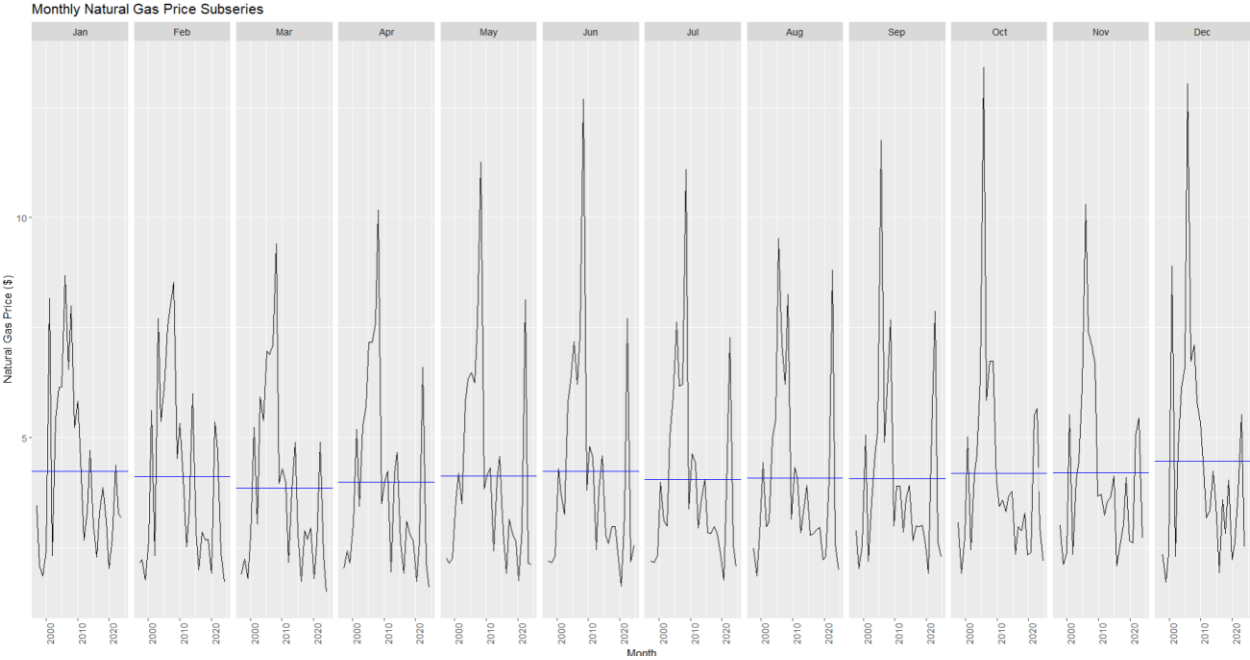


Figure 5: Classical Decomposition of Monthly Natural Gas Prices

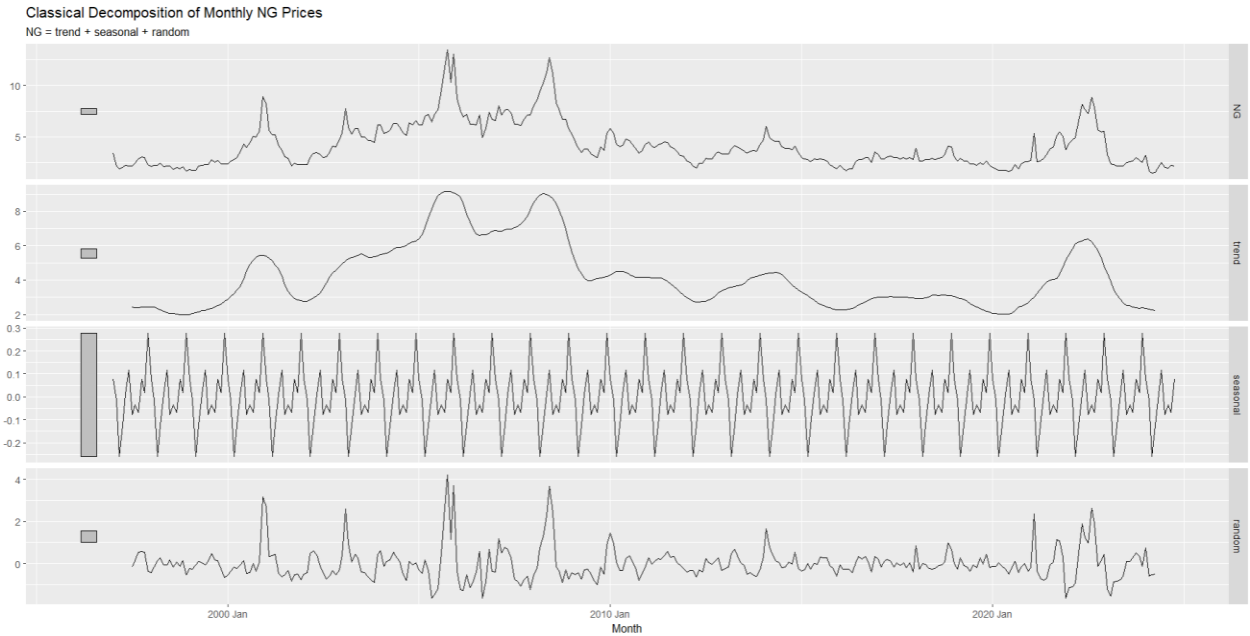


Figure 6: ACF of Detrended Monthly Natural Gas Prices: Classical Decomposition

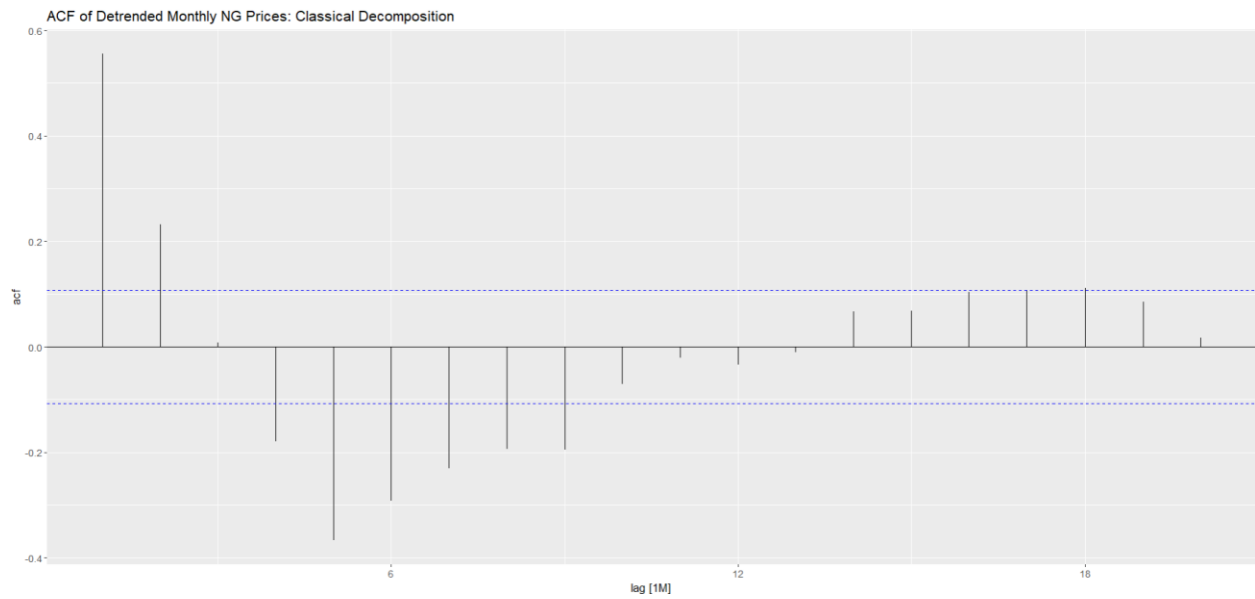


Figure 7: X11 Decomposition of Natural Gas Prices

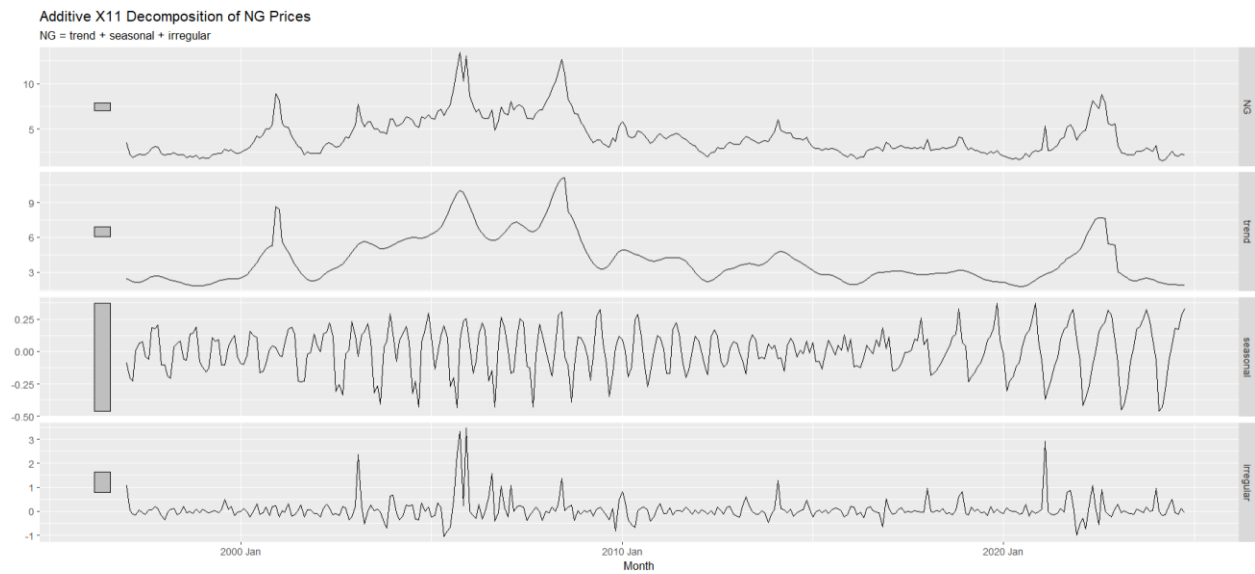


Figure 8: ACF of Residuals of Natural Gas Prices: X11 Decomposition

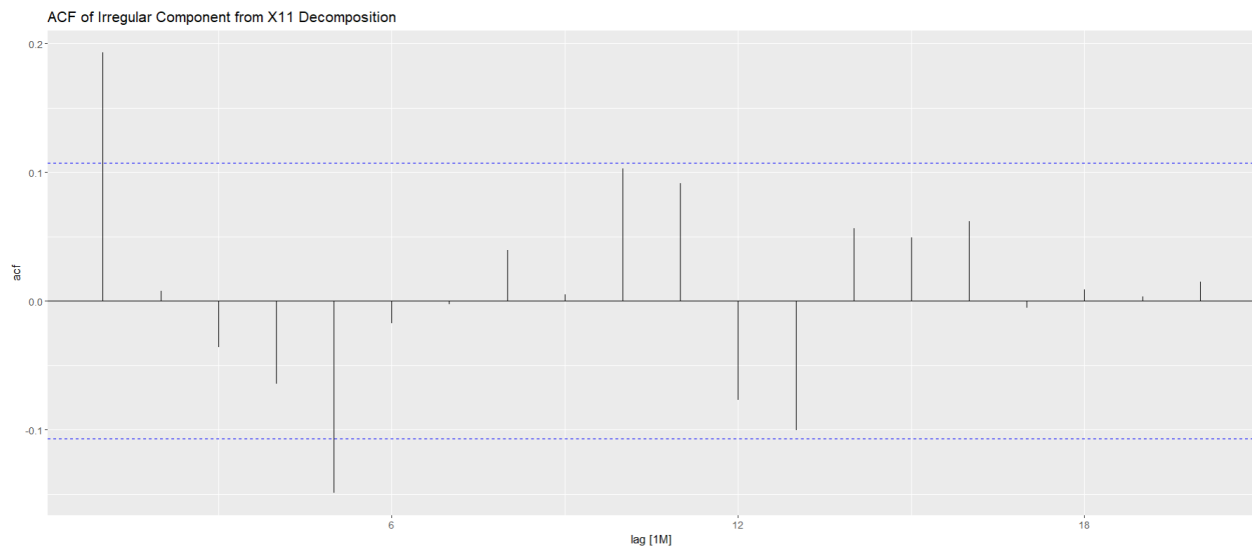


Figure 9: STL Decomposition of Natural Gas Prices

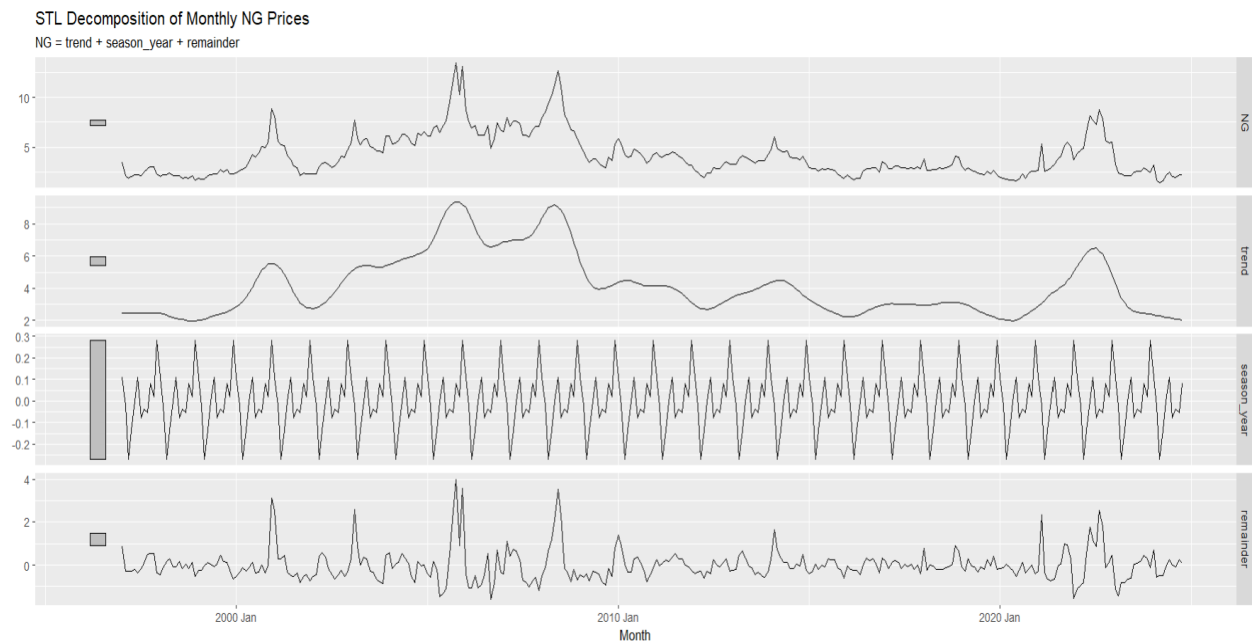


Figure 10: ACF of Residuals of Natural Gas Prices: STL Decomposition

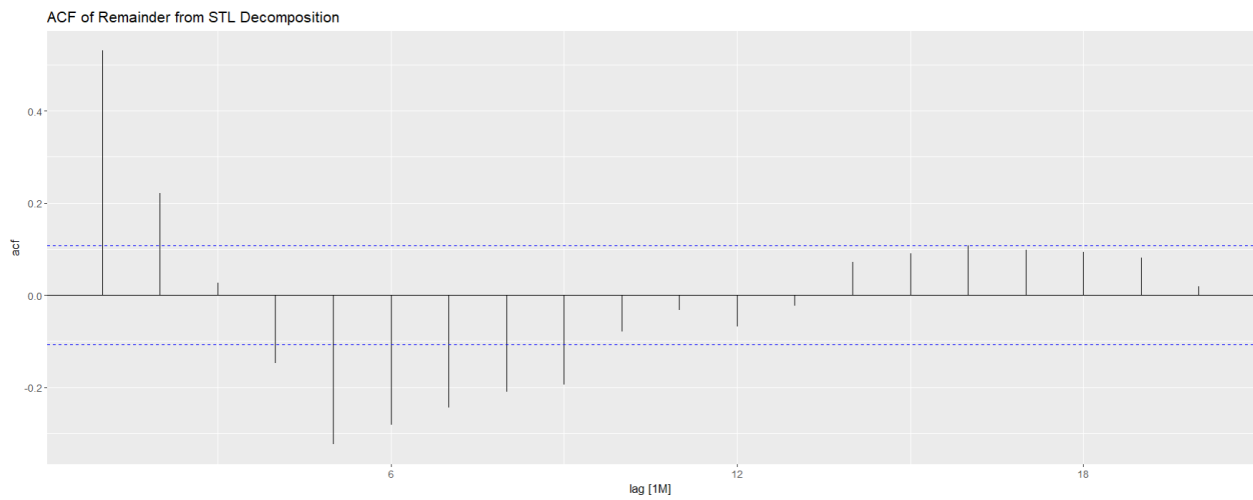


Figure 11: NG Price Decomposition using 3-month Moving Average

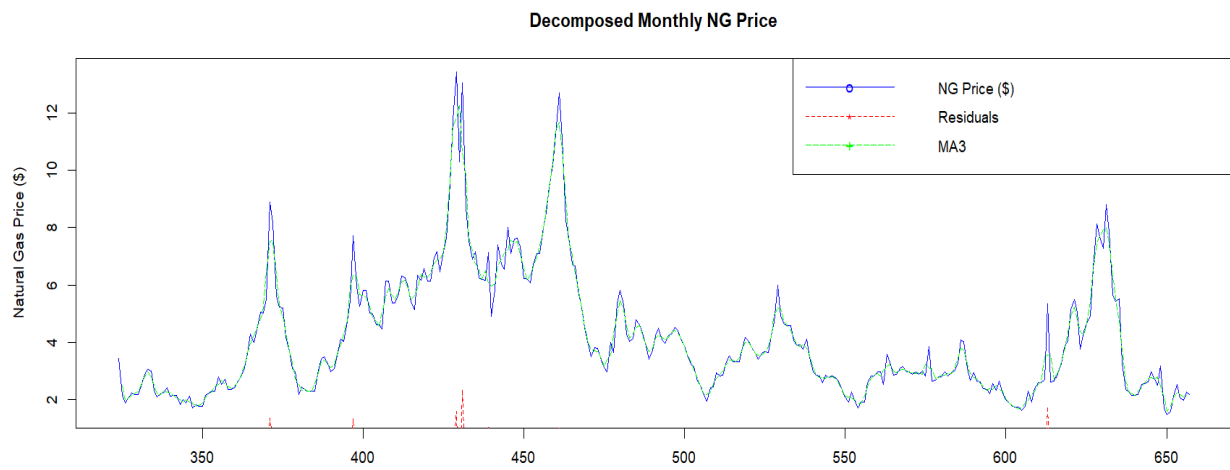


Figure 12: NG Prices and 3-month Moving Average Fit

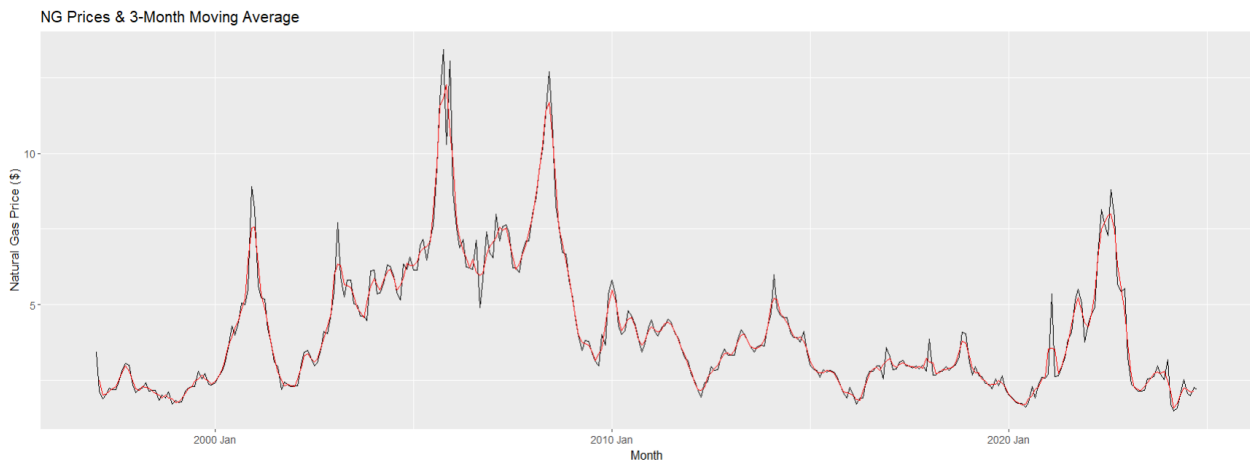


Figure 13: Log-transformed detrending of NG Prices Using Linear Regression

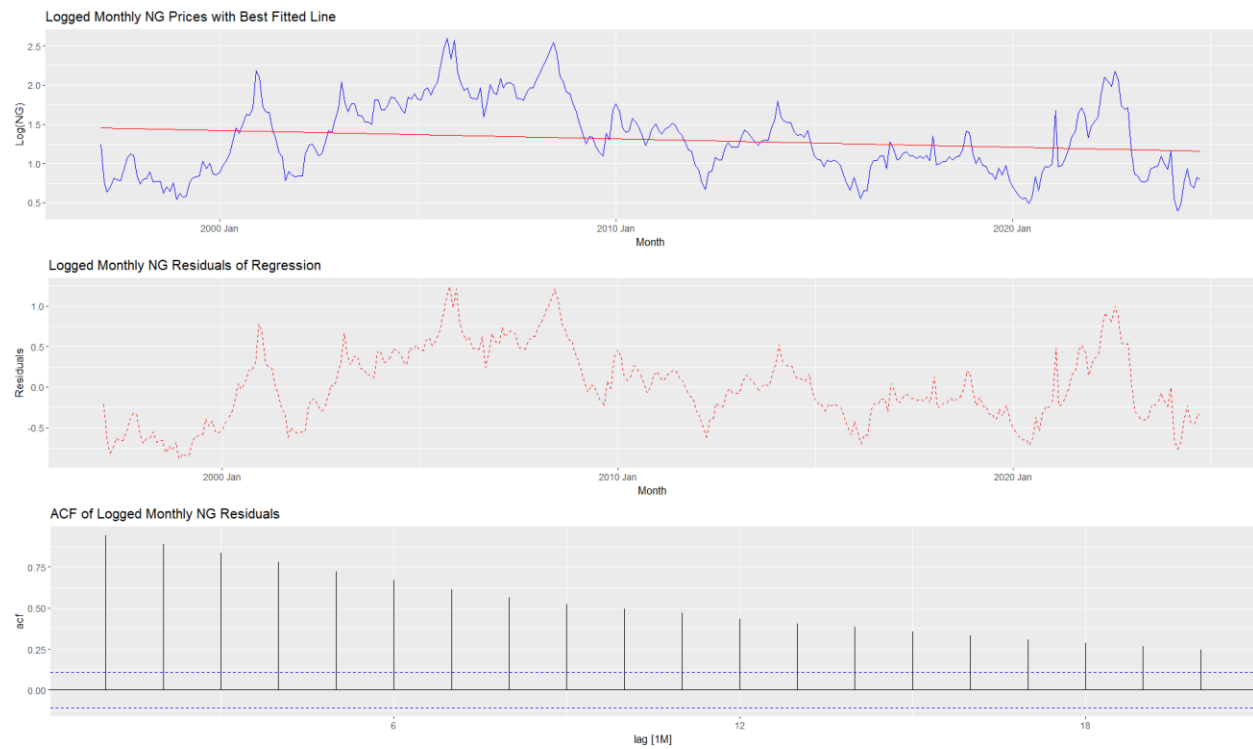


Figure 14: Detrending Using First Differencing of NG Prices

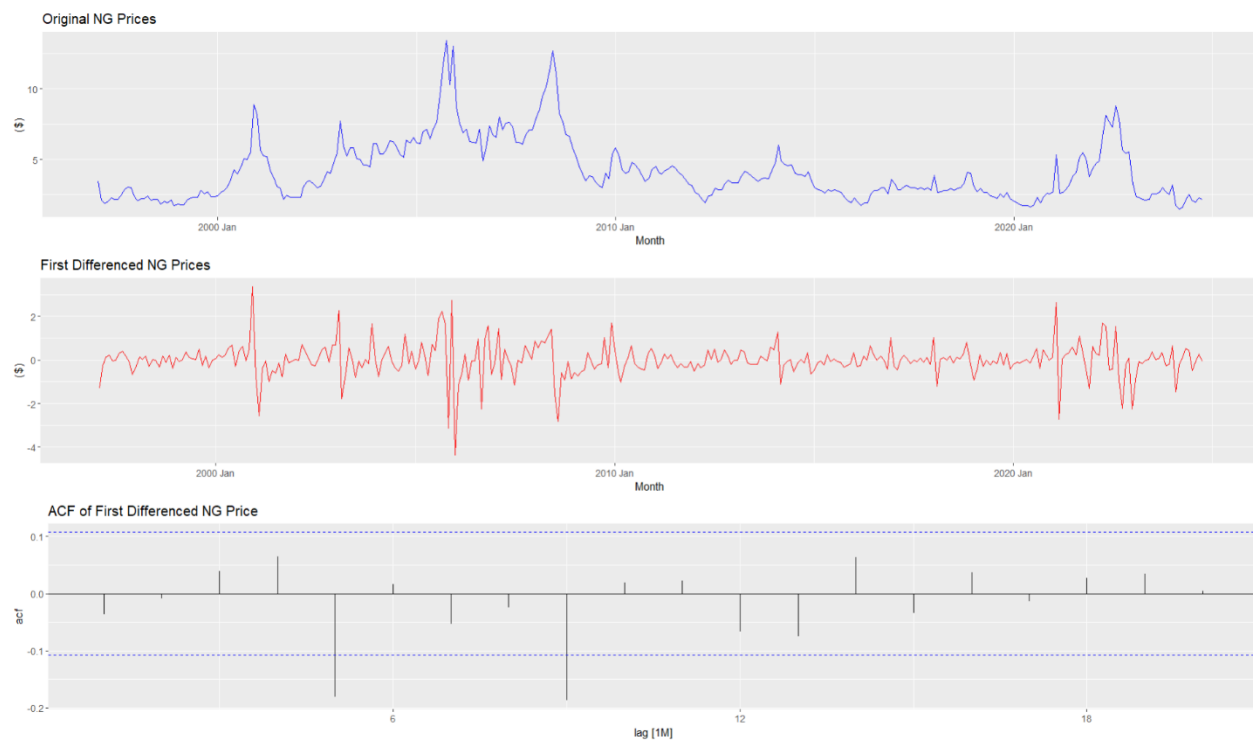


Figure 15: ACF of NG Prices after removing Seasonality Using Regression on Dummies

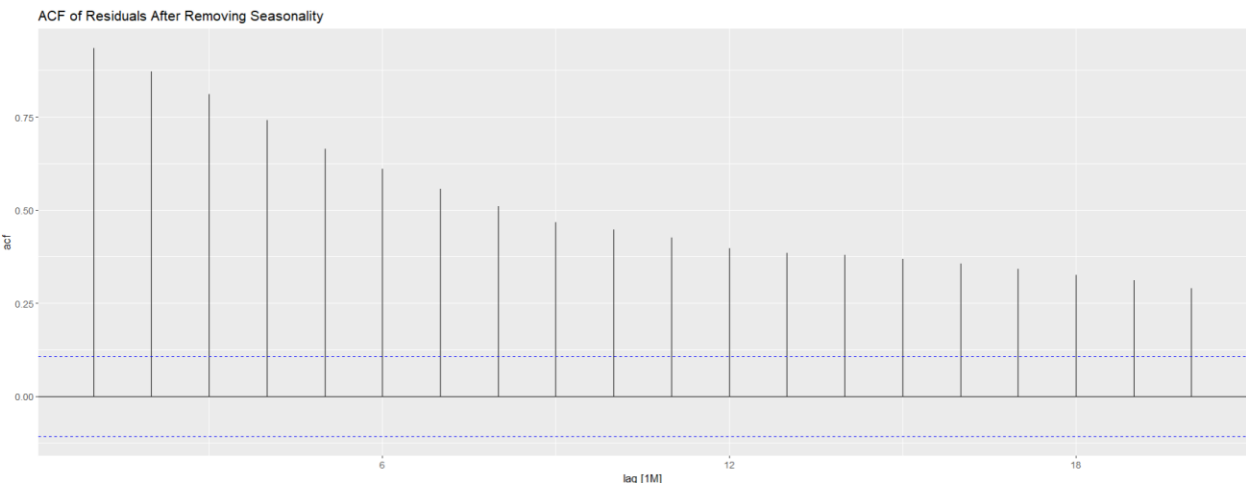


Figure 16: Residuals from Trend + Seasonality using TSML model for NG Prices

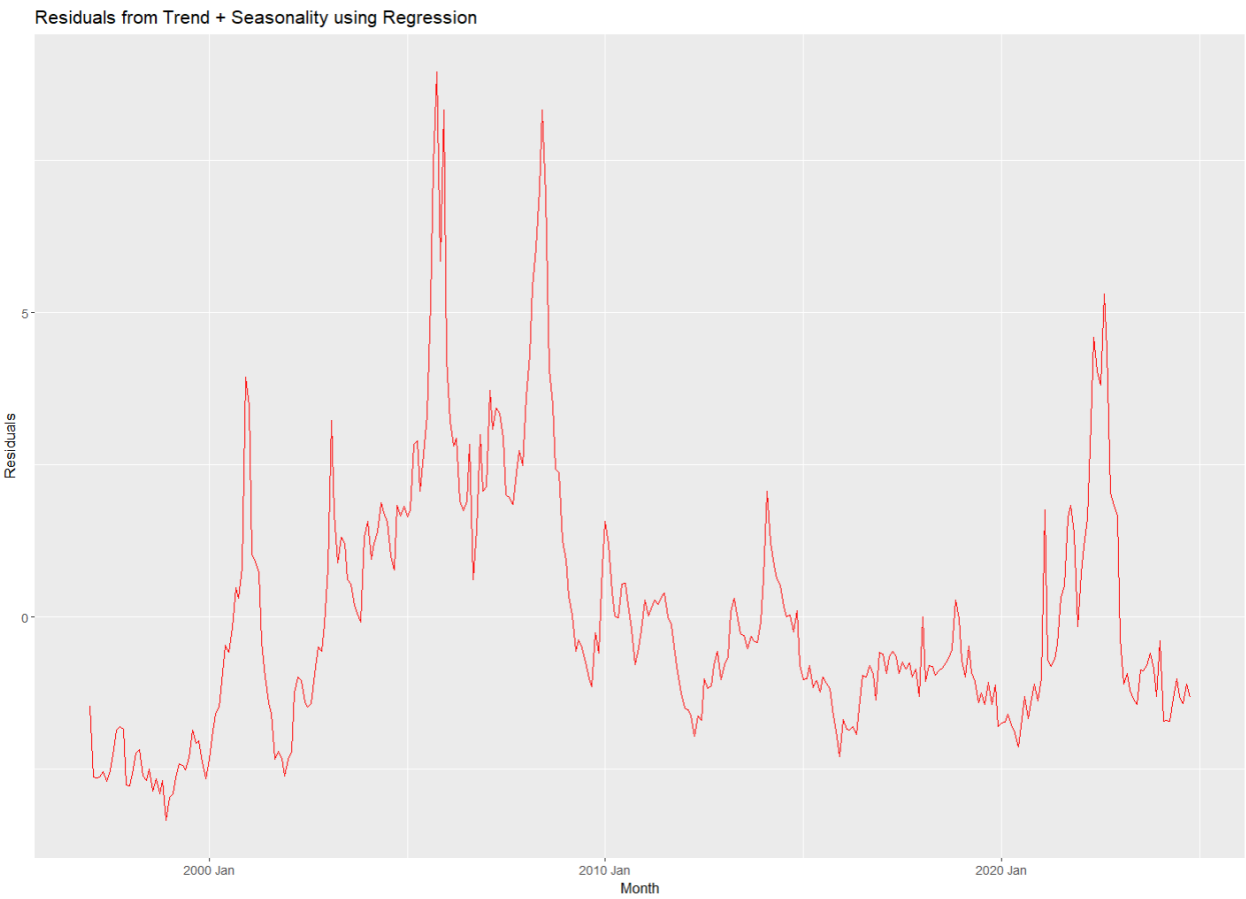


Figure 17: ACF of Seasonally Differenced NG Prices

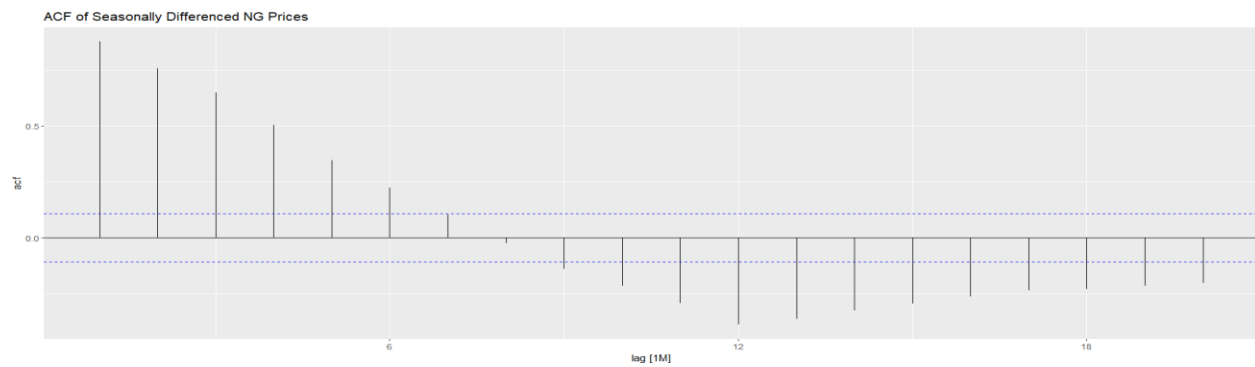


Figure 18: Residuals from Trend + Seasonality using Seasonal Difference of NG Prices

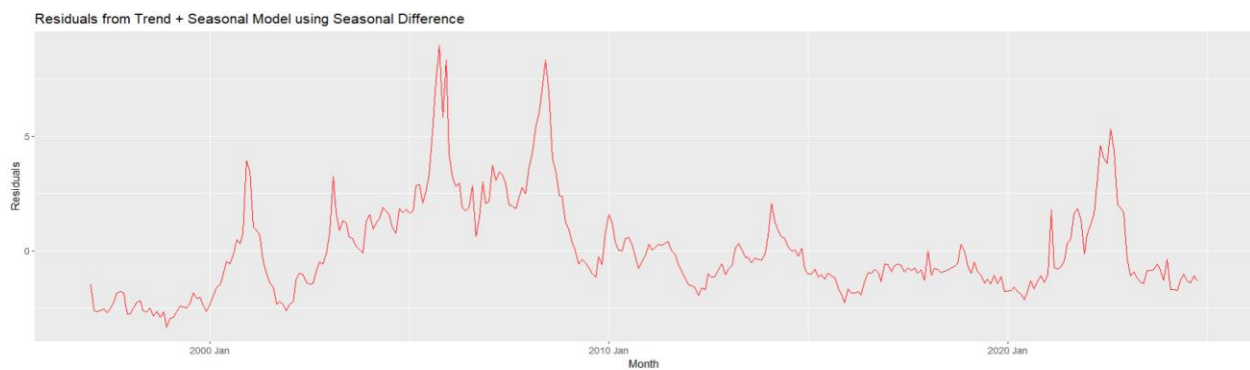


Figure 19: ACF of Sequentially Differenced Trend and Seasonality of NG Prices

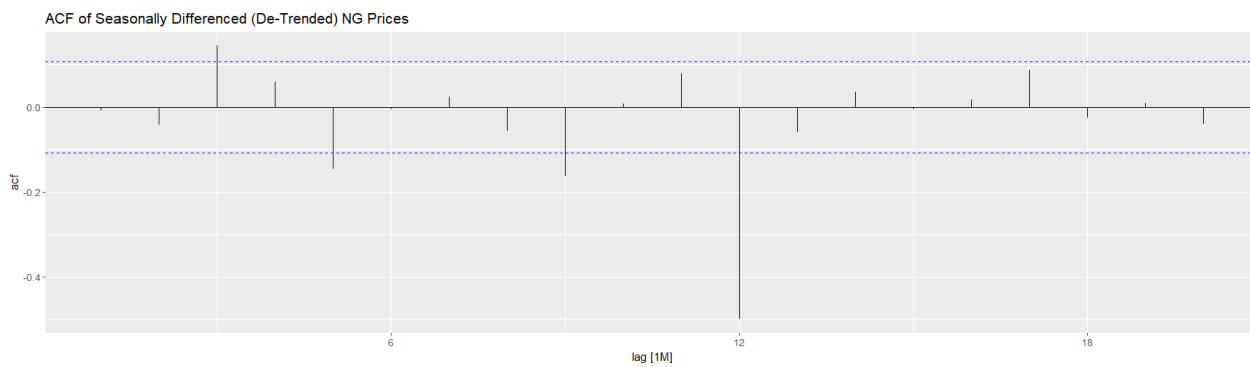
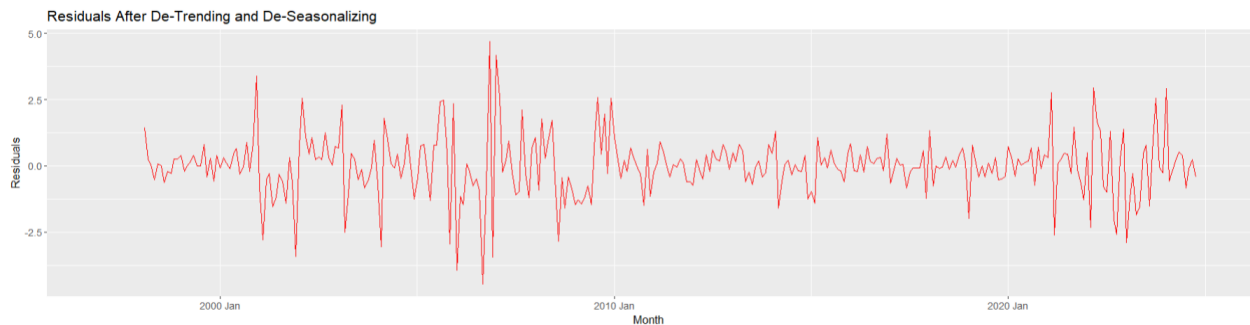


Figure 20: Residuals from De-trending and De-seasonalizing of NG Prices



The chart, titled "NG Price Forecasts Using Best ETS Model", plots the Natural Gas Price in dollars against time from 2000 to 2020. The y-axis, labeled "Natural Gas Price (\$)", ranges from 0 to 10. The x-axis, labeled "Month", shows major ticks for January 2000, January 2010, and January 2020. The historical data shows significant volatility, with major peaks around 2002, 2008 (reaching nearly \$12), and 2014. From 2020, the chart transitions into a forecast, represented by a shaded area indicating the 80% and 95% confidence levels. The forecast shows a peak around mid-2020 followed by a decline and then a slight recovery.

Model	ME	RMSE	MAE	MPE	MAPE	MASE	RMSSE	ACF1
ETS (M, N, N)	-3.18	3.21	3.18	-145	145	NaN	NaN	0.38
AIC = 1568.254, AICc = 1568.327, BIC = 1579.687								

Model	ME	RMSE	MAE	MPE	MAPE	MASE	RMSSE	ACF1
ARIMA	-3.18	3.21	3.18	-145	145	NaN	NaN	0.38
AIC = 768.93, AICc = 769.28, BIC = 795.59, lb stat 17, lb p-value 0.254.								

	ar1	ar2	ar3	ar4	ar5	sma1
Estimates	-0.0320	0.0021	0.0208	0.0599	0.1835	0.0904
S.E.	0.0541	0.0539	0.0555	0.0537	0.0540	0.0602

AIC = 768.93, AICc = 769.28, BIC = 795.59, lb stat 17, lb p-value 0.254.

Figure 22: Residual Diagnostic for ARIMA (5, 1, 0) (0, 0, 1) [12] model for NG Prices

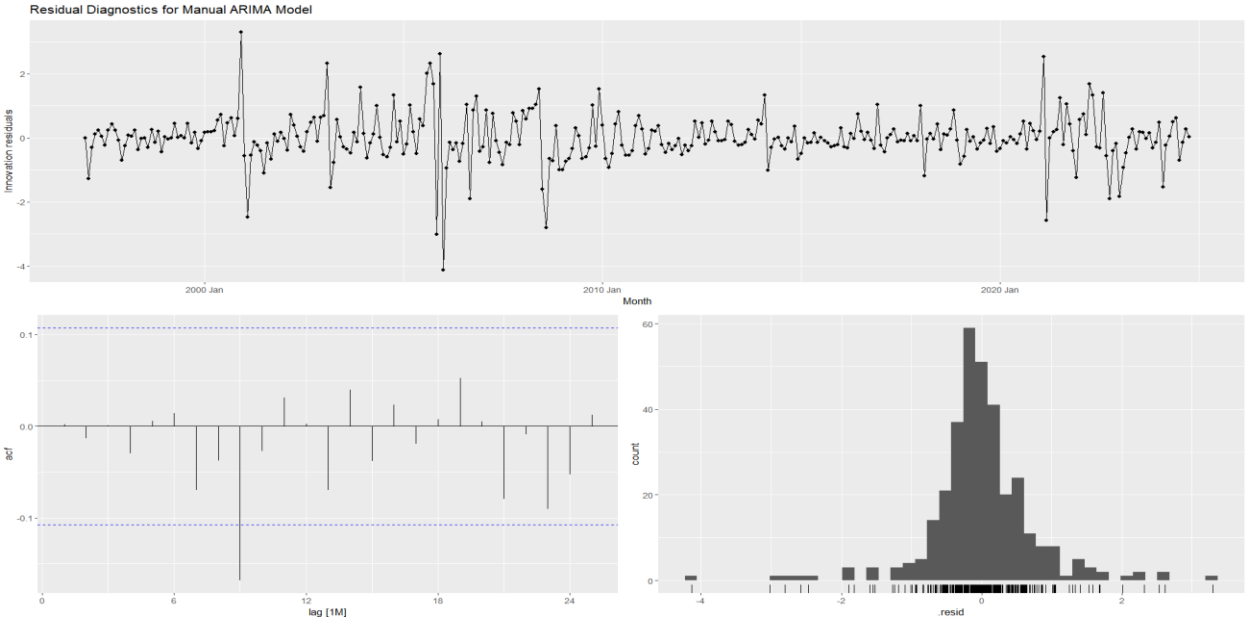


Figure 23: Test Set Forecast for NG Prices Using the Chosen ARIMA Model

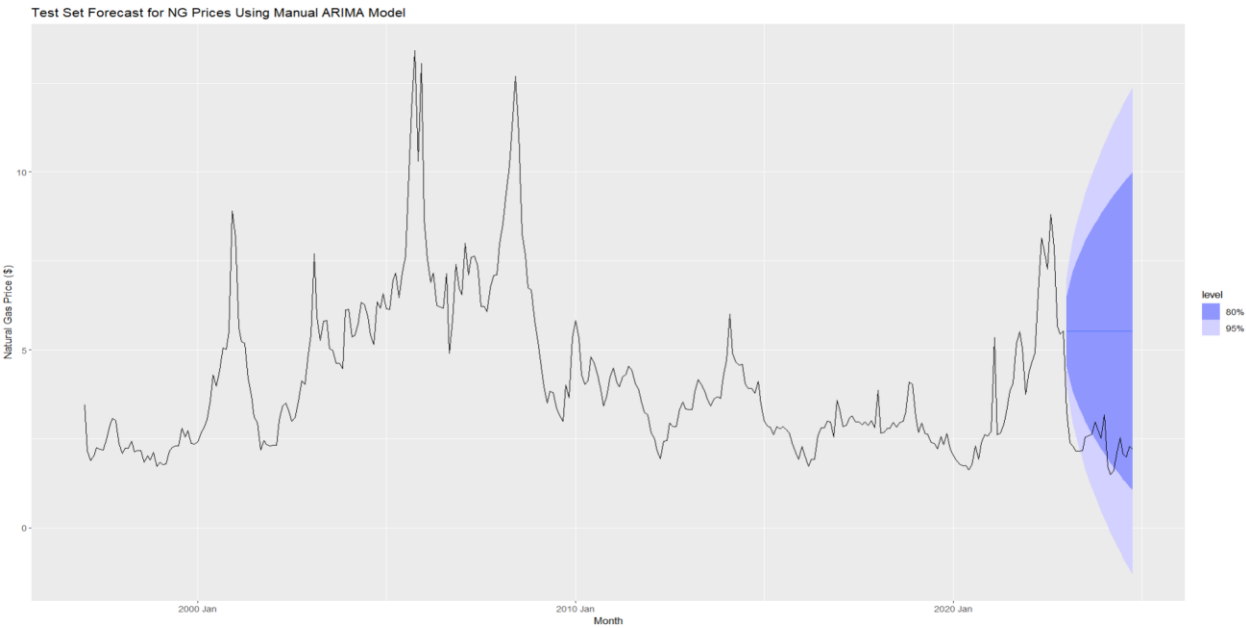


Figure 24: ARIMA (5, 1, 0) (0, 0, 1) [12] Model Forecast for NG Prices

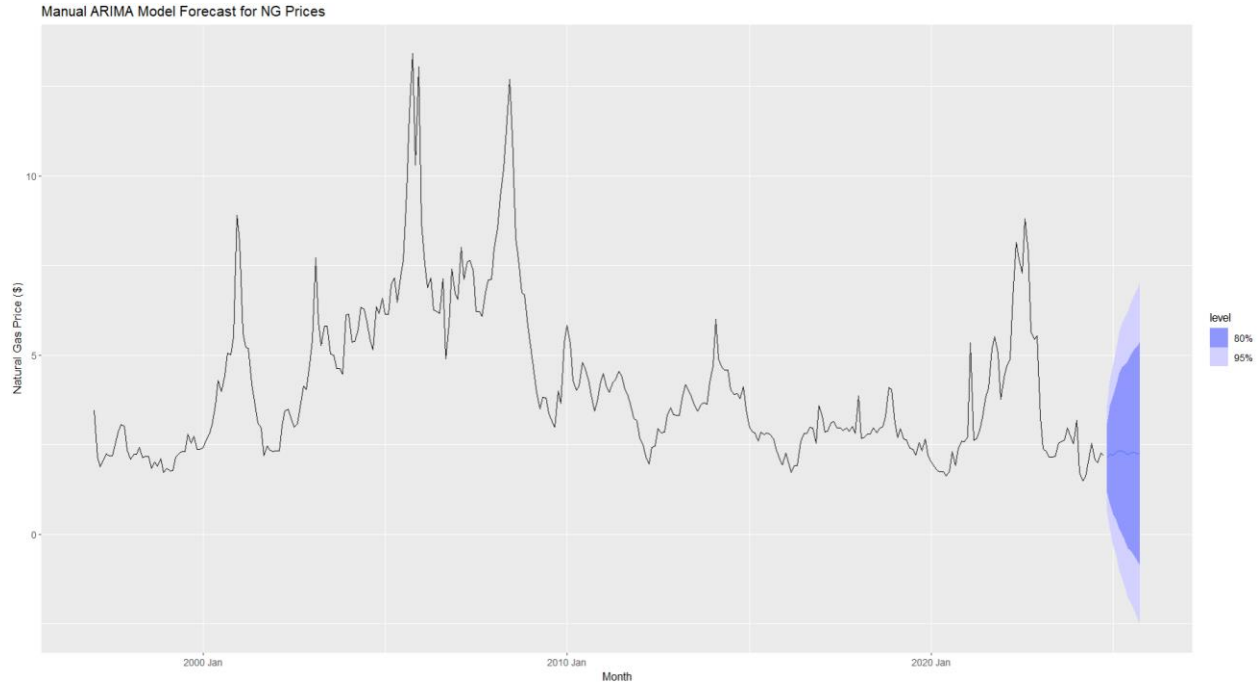


Table 6: Forecast Prices of NG with the Final ARIMA (5, 1, 0) (0, 0, 1) [12] Model

Month	Forecast Price (\$)
November 2024	2.12
December 2024	2.23
January 2025	2.21
February 2025	2.29
March 2025	2.32
April 2025	2.34
May 2025	2.27
June 2025	2.23
July 2025	2.28
August 2025	2.28
September 2025	2.25
October 2025	2.26

Table 7: Dynamic Regression with ARIMA (5, 1, 0) (0, 0, 1) [12] and Crude Oil Coefficients

	ar1	ar2	ar3	ar4	ar5	sma1
Estimates	-0.0320	0.0021	0.0208	0.0599	0.1835	0.0904
S.E.	0.0541	0.0539	0.0555	0.0537	0.0540	0.0602

lb stat 19.3, lb p-value 0.152.

Figure 25: Dynamic Regression Model Forecasts (ARIMA (5,1,0) (0,0,1) [12]) for NG

